

Risk and Opportunity Matrix – Sprint 1

Introduction

In Sprint 1, the group developed the initial risk and opportunity matrix for the Bayer Decision Agriculture project.

This stage focuses on mapping the primary uncertainties regarding the analysis of the ~320k rows dataset, the integration of public APIs (INMET/NOAA), the business problem framing, and the macro-environmental analysis (PESTEL). The risk assessment aligns with the project deliverables and the specific task distribution among the team members.

The matrix below presents the identified risks and opportunities, the fundamental justifications for their ratings, and the concrete action plans established to control them throughout the sprint.

Risks

Risk	Pro b. (%)	Impact	Grade / Intensi ty	Fundamenta l Justification	Risk Respon se	Action Plan (Summar y)	Owner / Functi on	Date Identifie d
Inconsisten cies in the core dataset (R1)	60%	Very High	Critical	The ~320k rows dataset may contain missing values or outliers in crucial variables (e.g., singulation, CV%), which directly compromises the simulator's logic.	Mitigate	Execute strict EDA to identify nulls and establish clear statistical dropping or imputation rules.	Arthur Almeid a & João Cauê / Data Analysi s	29/Apr/20 26

API rate limits or downtime (R2)	40%	High	Elevated	Relying on external public sources (INMET/Sentinel) introduces a technical dependency on third-party uptime for the simulator's inputs.	Mitigate	Download a static historical sample to serve as a local cache/mock backend during the development phases.	Yan Kruzicki / Risk Matrix	29/Apr/2026
Misalignment in Problem Framing (R3)	30%	Very High	Critical	Framing the wrong decision variables could render the final simulator impractical for Bayer's Farmers Insights Squad.	Mitigate	Validate the specific decision thresholds directly with Bayer stakeholders during the initial alignment workshop.	João Agmon / Problem Framing	29/Apr/2026
Superficial PESTEL Analysis (R4)	40%	Moderate	Relevant	Delivering generic macro factors without mathematically connecting them to the ROI of the planting strategy reduces the	Avoid	Cross-review the PESTEL factors with the Framing team to ensure all points tie directly to the	Rafael Skitch / PESTEL	29/Apr/2026

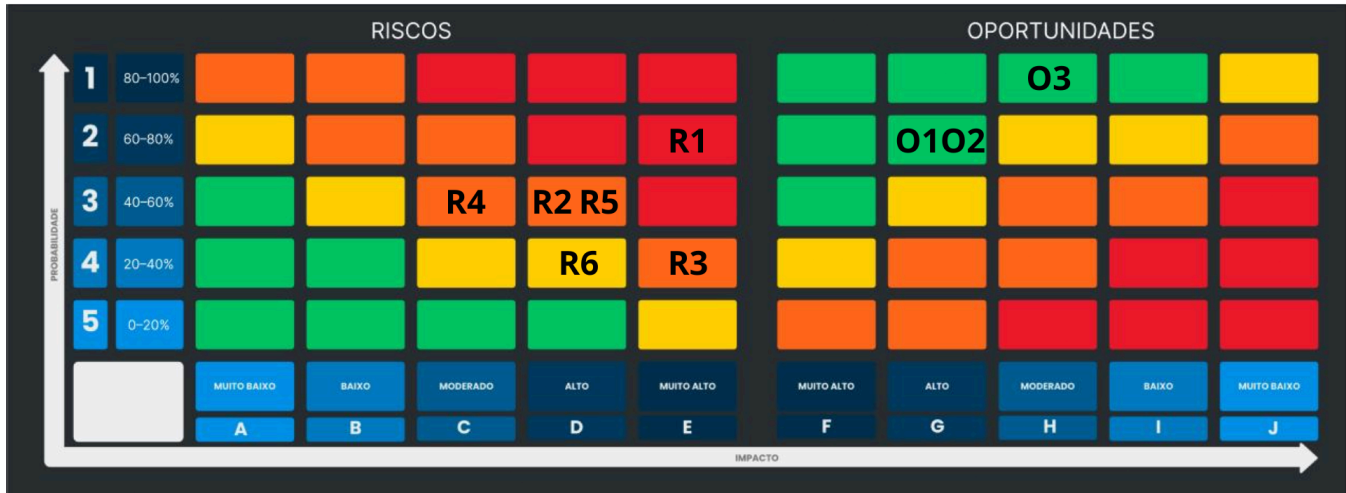
				artifact's utility.		project scope.		
Retroactive GitLab management penalties (R5)	50%	High	Elevated	Inteli's evaluation strictly penalizes retroactive issue tracking, directly lowering the individual performance factor score.	Mitigate	Implement a daily check to ensure all artifact tasks are updated in the "Doing" and "Testing" columns.	Miguel Almeida / PESTEL	29/Apr/2026
Scope creep toward legacy systems (R6)	20%	High	Relevant	The team might waste resources attempting to integrate the prototype with Bayer's internal systems, which is expressly out of scope.	Avoid	Explicitly document the out-of-scope boundaries within the Problem Framing deliverable.	João Agmont / Problem Framing	29/Apr/2026

Opportunities

Opportunity	Prob. (%)	Impact	Grade / Intensity	Fundamental Justification	Opportunity Response	Action Plan (Summary)	Owner / Function	Date Identified

High correlation findings in EDA (O1)	70%	High	Elevated	Finding strong statistical correlations early between public weather data and planting efficiency creates a solid base for the decision tree.	Exploit	Highlight specific variable correlations in the EDA report to directly guide the logic of Sprint 3.	João Cauê / Data Analysis	29/Apr/2026
Strategic differentiation via PESTEL(O2)	60%	High	Elevated	A deep, quantified analysis of Brazilian agribusiness volatility provides a unique angle for the subsequent Payoff Matrix.	Enhance	Quantify market sizes and regulatory impacts within the PESTEL document instead of just listing qualitative factors.	Ariel Mouadeb / PESTEL	29/Apr/2026

Establishing a reusable Risk Framework (O3)	80%	Moderate	Relevant	Building a highly structured matrix now accelerates the mandatory update process required in all subsequent sprints.	Exploit	Keep the matrix mapping directly tied to the workflow, allowing rapid status updates for Sprint 2.	Yan Kruziski / Risk Matrix	29/Apr/2026
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Conclusion

Sprint 1 establishes the analytical and strategic foundation for the decision-making simulator. The primary risks identified relate to data integrity within the EDA phase and the accuracy of the strategic alignment in the Problem Framing.

The application of action plans by the respective artifact owners ensures that technical dependencies (such as external APIs) and internal agile management (GitLab tracking) are controlled early. Parallely, the depth required in the PESTEL and EDA deliverables opens clear opportunities to strengthen the

mathematical and business logic that will be demanded in upcoming sprints. This matrix will act as a living document and will be reviewed at the beginning of Sprint 2.